

MLOps Churn Project - Execution Guide (Detailed)

1. Environment Setup

Create an isolated environment to avoid dependency conflicts.

```
python -m venv venv
venv\Scripts\activate
pip install -r requirements.txt
```

2. Generate Dataset

Generate synthetic churn dataset with probabilistic behavior.

```
python data/generate_data.py
```

3. Train the Model

Train models and log results in MLflow.

```
python src/train.py
```

Inside train.py, models are defined like:

```
modelos = {
    'RandomForest': RandomForestClassifier(n_estimators=200),
    'LogisticRegression': LogisticRegression(max_iter=1000)
}
```

4. Start MLflow UI

Visualize experiments and metrics.

```
mlflow ui
# Access http://127.0.0.1:5000
```

5. Run the API

Start API to serve predictions.

```
uvicorn deployment.api:app --reload
# Access http://127.0.0.1:8000/docs
```

6. Test Prediction

Send a POST request with JSON input:

```
{
  "tenure": 10,
  "MonthlyCharges": 80,
```

```
"TotalCharges": 500,  
"Contract": "Month-to-month",  
"InternetService": "Fiber optic"  
}
```

7. Model Parameter Tuning

Edit parameters inside src/train.py

RandomForest example:

```
RandomForestClassifier(n_estimators=200, max_depth=10, min_samples_leaf=5)
```

Key parameters:

- n_estimators: number of trees - max_depth: controls complexity - min_samples_leaf: reduces overfitting

Logistic Regression example:

```
LogisticRegression(C=1.0, solver='liblinear')
```

Key parameters:

- C: regularization strength - solver: optimization algorithm

Threshold tuning example:

```
y_pred = (y_prob >= 0.5)  
# change to:  
y_pred = (y_prob >= 0.3) # more recall  
y_pred = (y_prob >= 0.7) # more precision
```

8. Notes

Best practices:

- Register model before API - Use latest version - Do not commit venv/ or mlruns/